

Education

M.S. in Computer Science

May 2025

B.S. in Computer Science

May 2024

Work Experience

Junior Software Developer

May 2024– Present

- Built and launched a production .NET + React Website supporting 500-1000 daily users, engineering the full stack from backend APIs to frontend architecture.
- Improved page load from 2.4s → 1.0s and boosted Lighthouse performance scores by +38 points through systematic performance tuning.
- Engineered high-performance frontend architecture using React Context, Redux and TanStack Query, simplifying state management and reducing unnecessary renders by ~30% and significantly improving UI time-to-interactive across mission-critical workflows.
- Designed and deployed PL/SQL procedures powering 5+ microservices, reducing data-related defects and enabling consistent validation across distributed systems.
- Delivered a .NET Core background service integrated with Azure Service Bus, achieving 100% automated nightly data synchronization and eliminating manual operational tasks.
- Reduced backend load up to ~60% and improved API response times by ~35% through Redis caching, query optimization and restructuring of high-throughput data paths.
- Containerized backend services with Docker and deployed to Azure, eliminating environment drift and reducing deployment setup time from hours to minutes.
- Built CI/CD pipelines using GitHub Actions and Jenkins, automating builds, testing and security scans, reducing release cycle time by ~40% and increasing deployment reliability.
- Managed IIS production deployments, SSL certificate installation, and environment configuration improving up time and reducing deployment failure rates.
- Strengthened application security by remediating high severity vulnerabilities in Veracode, achieving 100% security compliance and preventing recurring exposure.
- Increased automated test coverage by ~30% using NUnit, Selenium, and Postman accelerating defect detection.
- Unified logging using Serilog and Kibana, enhancing production debugging and visibility.

Skills

- **Languages:** Python, TypeScript, JavaScript, C#, C/C++, Java, Bash, Swift
- **Frontend:** React, Next.js, Redux, TanStack Query, TailwindCSS, Bootstrap, HTML, CSS
- **Backend:** ASP.NET Core, Node.js, Flask, REST APIs, Microservices
- **Cloud:** AWS (EC2, Lambda, S3, RDS, API-Gateway, SQS, IoT Greengrass), Azure
- **Databases:** Oracle PL/SQL, PostgreSQL, MySQL, SQL Server, DynamoDB, MongoDB, Redis
- **DevOps:** Docker, GitHub Actions, Jenkins, CI/CD, IIS, SSL
- **Testing and QA:** NUnit, Selenium, Postman, Veracode Jenkins, Serilog, Kibana
- **Awards & Honors:** 5x Hackathon Winner

Project Experience

Face Recognition on the Edge

Mar 2025 - May 2025

Class Project

Python, AWS IoT Greengrass, MQTT, Amazon SQS, AWS Lambda

- Designed and implemented a distributed edge to cloud face recognition pipeline using AWS IoT Greengrass Core on EC2 for edge inference and AWS Lambda for processing on cloud via SQS.
- Engineered MQTT protocol communication and automated deployment of Greengrass components, enabling reproducible and scalable edge environments.
- Benchmarked the system on 1,000+ frames, achieving 100% classification accuracy with ~1s end-to-end latency.